

LESSON 8-1 • SESSION 1 • PRACTICE • FORM A

Equations and Solutions

Learning target – I can check whether a value is a solution of an equation by substituting it. · 6.EE.B.5

REMEMBER

- An **equation** says two amounts are equal: $x + 3 = 10$.
- To test a value, **substitute** it for the variable. Then do the math.
- If both sides match, the value is a **solution**. $4k = 20$ with $k = 5$: $4 \times 5 = 20$, true.

WORKED EXAMPLE

A basket has b apples. You add 9 more and have 15. Is $b = 6$ a solution of $b + 9 = 15$?

Step 1 Substitute: replace b with 6 to get $6 + 9$.

Step 2 Add: $6 + 9 = 15$.

Step 3 Compare: $15 = 15$, so the equation is true.

Answer Yes, $b = 6$ is a solution.

WORDS equation (*ecuación*) · solution (*solución*) · substitute (*sustituir*)

PRACTICE

show your work · label units

1. Substitute the value. Write **T** for true or **F** for false.

a. $s - 15 = 25$ ($s = 40$) ____

b. $7r = 42$ ($r = 5$) ____

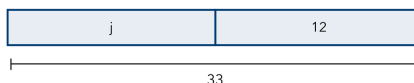
c. $p + 18 = 47$ ($p = 29$) ____

d. $4k = 36$ ($k = 32$) ____

e. $n - 4 = 3$ ($n = 7$) ____

f. $m + 23 = 41$ ($m = 18$) ____

2. **Juice Box Cooler.** A cooler holds j juice boxes. The coach adds 12 more. Now there are 33. The tape diagram shows $j + 12 = 33$.



a. Test $j = 21$. ____

b. Test $j = 19$. ____

c. Which value is the solution? ____

3. **Pencil Packs.** Tanisha wrote $6p = 72$ for packs of pencils. Find the solution. Prove it.

Show your work.

$p =$ _____

4. **Bake Sale.** Amara wrote $c - 24 = 36$ for cupcakes left. Is $c = 60$ a solution? Check.

Hint Replace c with 60. Do both sides match?

Show your work.

Yes or no _____

5. **Equal Teams.** Coach writes $4t = 28$ for equal teams. Ray says $t = 7$. Is Ray correct? Check.

Hint $4t$ means $4 \times t$.

Show your work.

Yes or no _____

6. **Library Fine.** Jun owes a library fine. After he pays \$2, he still owes \$11. Write an equation using f for the starting fine. Then find f by testing values.

Hint Start minus what he paid leaves what he owes.

Show your work.

Equation; f _____

7. **Game Tokens.** Deshawn wrote $5g = 45$ for the games he can play with 45 tokens. He says $g = 8$. Check his value. Then find the real solution.

Hint Substitute his value first. Then try the next number.

Show your work.

$g =$ _____

8. **Number Puzzle.** A puzzle reads: a number plus 17 equals 45.

- Write the equation using n . _____
- Test $n = 28$. Is it the solution? _____
- How is testing a value different from solving? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Maya's Work.** To check whether $k = 12$ is a solution of $k + 30 = 42$, Maya wrote $12 + 30 = 42$, so $42 = 42$. Then she said the solution is 42. Explain her mistake.

TEST PRACTICE

choose the best answer

10. Is $x = 6$ a solution of $x + 5 = 11$?

A yes **B** no **C** only if $x = 5$ **D** cannot tell

11. Is $y = 3$ a solution of $5y = 20$?

A yes **B** no **C** only if $y = 5$ **D** cannot tell

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

12 Q	45 Z	432 X	62 J	21 E	13 U	28 L	40 K	9 A
		#2c	#3	#6	#7	#8b		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____